

```

TITLE 'Soft Drink BottlingExperiment Example 5.3';
DATA BOTTLING;
INPUT PRESSURE SPEED CARBONATION REPLICATE FILLHEIGHT @@;
      CARDS;
25 200 10 1 -3  25 200 10 2 -1  25 200 12 1 0  25 200 12 2 1  25 200 14 1  5  25 200 14 2  4
25 250 10 1 -1  25 250 10 2  0  25 250 12 1 2  25 250 12 2 1  25 250 14 1  7  25 250 14 2  6
30 200 10 1 -1  30 200 10 2  0  30 200 12 1 2  30 200 12 2 3  30 200 14 1  7  30 200 14 2  9
30 250 10 1  1  30 250 10 2  1  30 250 12 1 6  30 250 12 2 5  30 250 14 1 10  30 250 14 2 11
;
DATA BOTTLING1;
SET BOTTLING;
TFILLHEIGHT=LOG(FILLHEIGHT+20);
PROC GLM PLOTS=(DIAGNOSTICS);
      CLASS PRESSURE SPEED CARBONATION REPLICATE;
      MODEL TFILLHEIGHT=PRESSURE|SPEED|CARBONATION;
      MEANS PRESSURE|SPEED|CARBONATION/TUKEY LINES;
      OUTPUT OUT=NEW P=PREDY R=RESID;
PROC UNIVARIATE NORMAL;
VAR RESID;
RUN;

```